

DS210 Final Project Proposal

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Dataset: Stanford Web Graph Dataset

(<https://snap.stanford.edu/data/web-Stanford.html>)

1. I plan to use the Stanford Web Graph dataset, which represents web pages and hyperlinks between them from the stanford.edu domain. Each node represents a web page and the edges represent hyperlinks. I'm interested in this dataset because it captures a real-world network with the opportunity to graph theoretical tools and algorithms to explore patterns and behaviors commonly observed in large-scale graphs.

2. I intend to examine the degree distribution – I will compute the in-degree and out-degree distributions. I want to particularly focus on understanding the distributions of node degrees in this graph and whether it follows a power-law distribution generally expected for many real-world networks. I'll evaluate the goodness of fit on the power-law using a simple regression and then provide my interpretations of the results. I'll likely begin this project by using a sample of the graph and then scale it up once my code works (and if my system or cloud environment is able to handle it).

3. The following is the plan I have to execute this project in a timely order.

Component #1 (Week of April 7th – April 13th):

- Receive feedback (if any modifications need to be made)
- Parse the dataset and build directed graph.
- Focus on a subset / sample of the graph (since this graph may be too huge to handle).

Component #2 (April 14th – April 22nd):

- Compute in-degree and out-degree for the nodes and store.
- Convert data into vectors.
- Build a simple regression model and perform goodness-of-fit.
- Extract slope and intercept + Compute R^2 for regression quality.
- Interpret power-law distribution results.

Component #3 (April 23rd – May 1st):

- Refactor code and prepare GitHub repository.
- Summarize results in write-up (either as separate PDF or within README.md)
- Submit project.